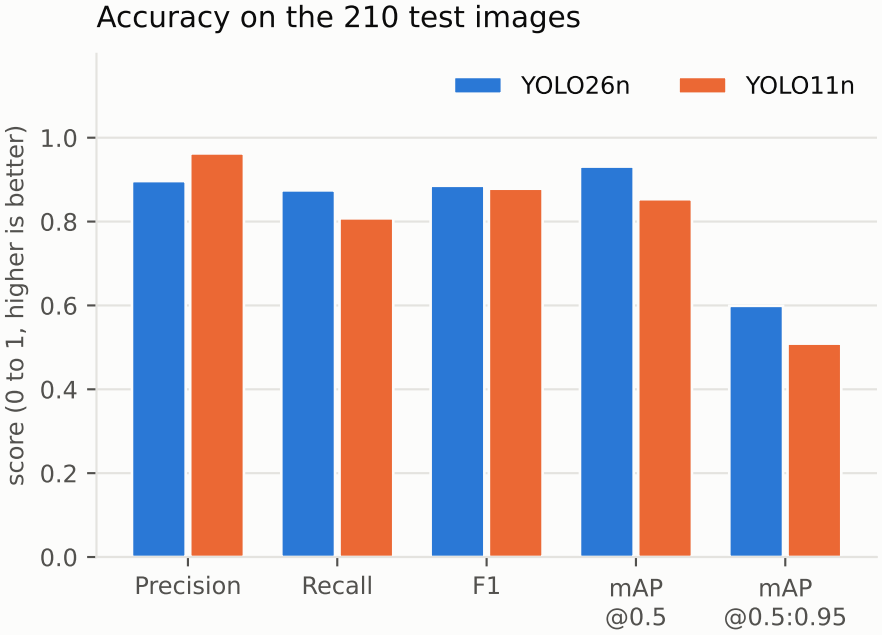
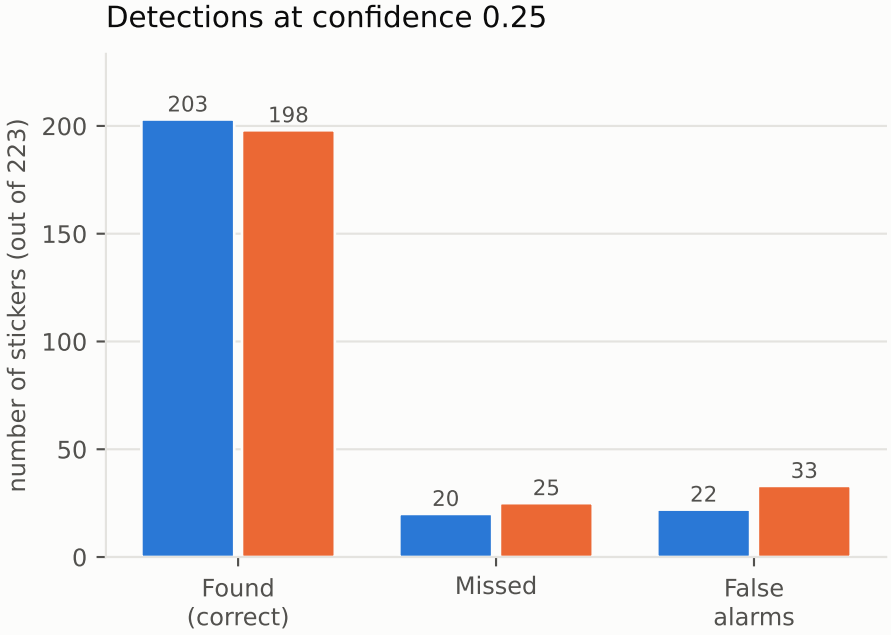
YOLO26n vs YOLO11n

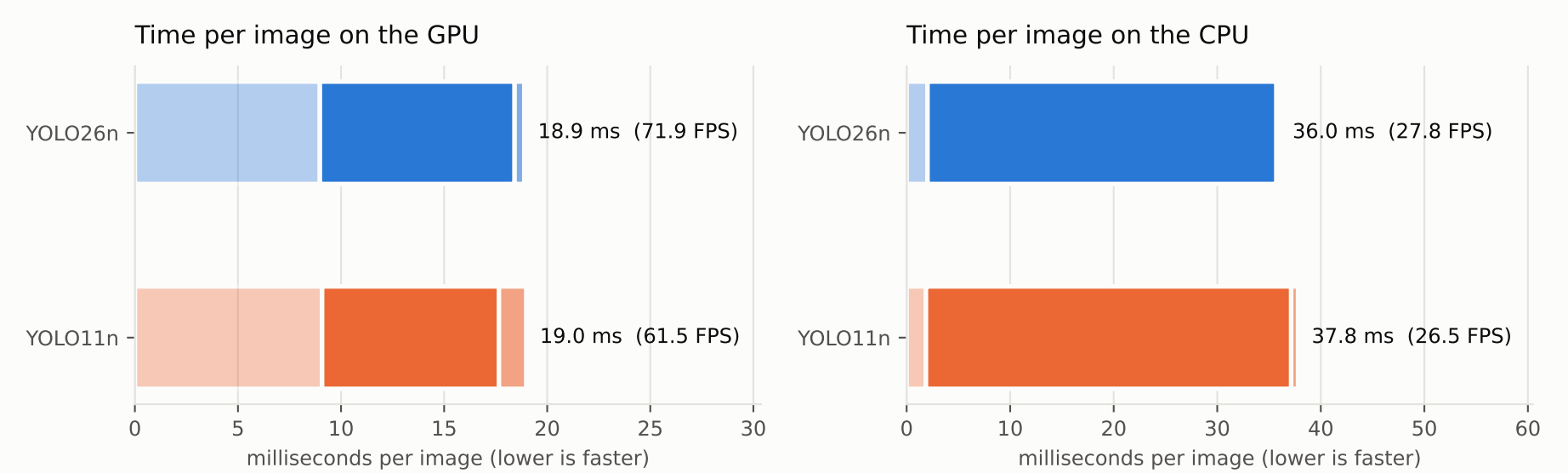
PSU Vehicle Sticker Detection Benchmark

November 13, 2026

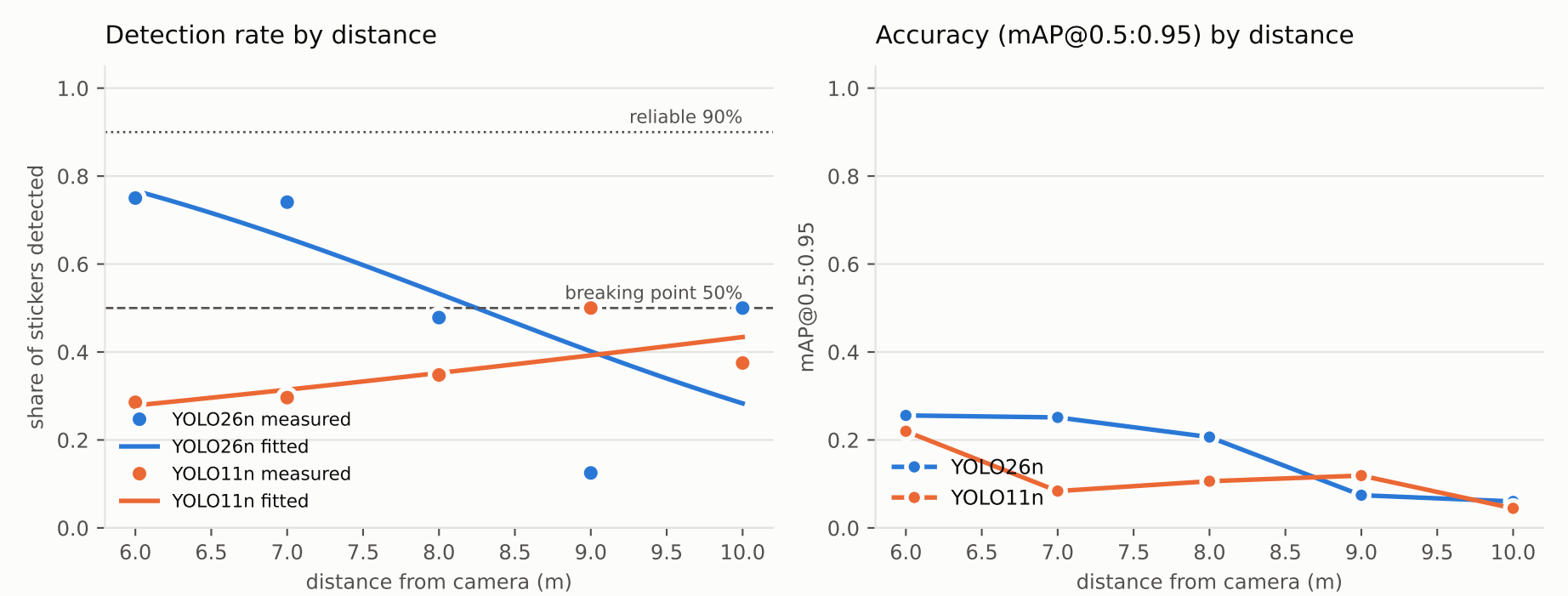
Measure	YOLO26n	YOLO11n	Better
Accuracy: mAP@0.5:0.95	0.599	0.509	YOLO26n
Accuracy: mAP@0.5	0.931	0.853	YOLO26n
Precision	0.897	0.963	YOLO11n
Recall	0.874	0.808	YOLO26n
Speed on GPU (FPS)	71.9	61.5	About the same
Speed on CPU (FPS)	27.8	26.5	YOLO26n
Distance: still detected up to (50%)	8.2 m	—	YOLO26n found more stickers (McNe
Motion blur: breaks at (50%)	64 px	71 px	YOLO11n
Occlusion: breaks at (50%)	46% covered	44% covered	YOLO26n
Low light: breaks at (50%)	gamma 3.83	gamma 4.10	YOLO11n
Glare: breaks at (50%)	gain 3.27x	gain 3.16x	No clear difference (YOLO26n slightl

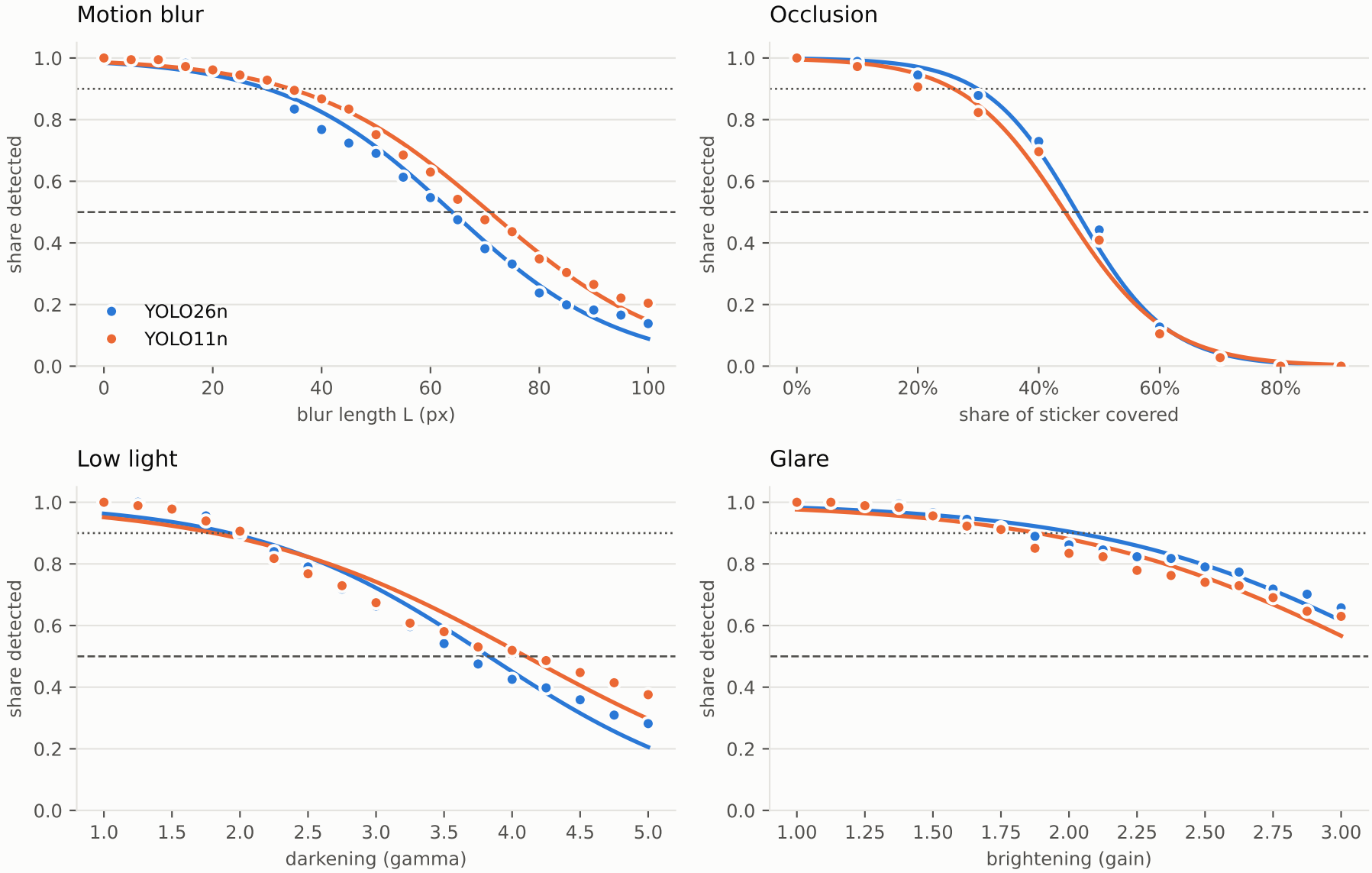

Accuracy on the 210 test images
YOLO26n
YOLO11n
score (0 to 1, higher is better)
1.0
0.8
0.6
0.4
0.2
0.0
Precision
Recall
F1
mAP @0.5
mAP @0.5:0.95


Detections at confidence 0.25
number of stickers (out of 223)
200
150
100
50
0
203
198
20
25
22
33
Found (correct)
Missed
False alarms

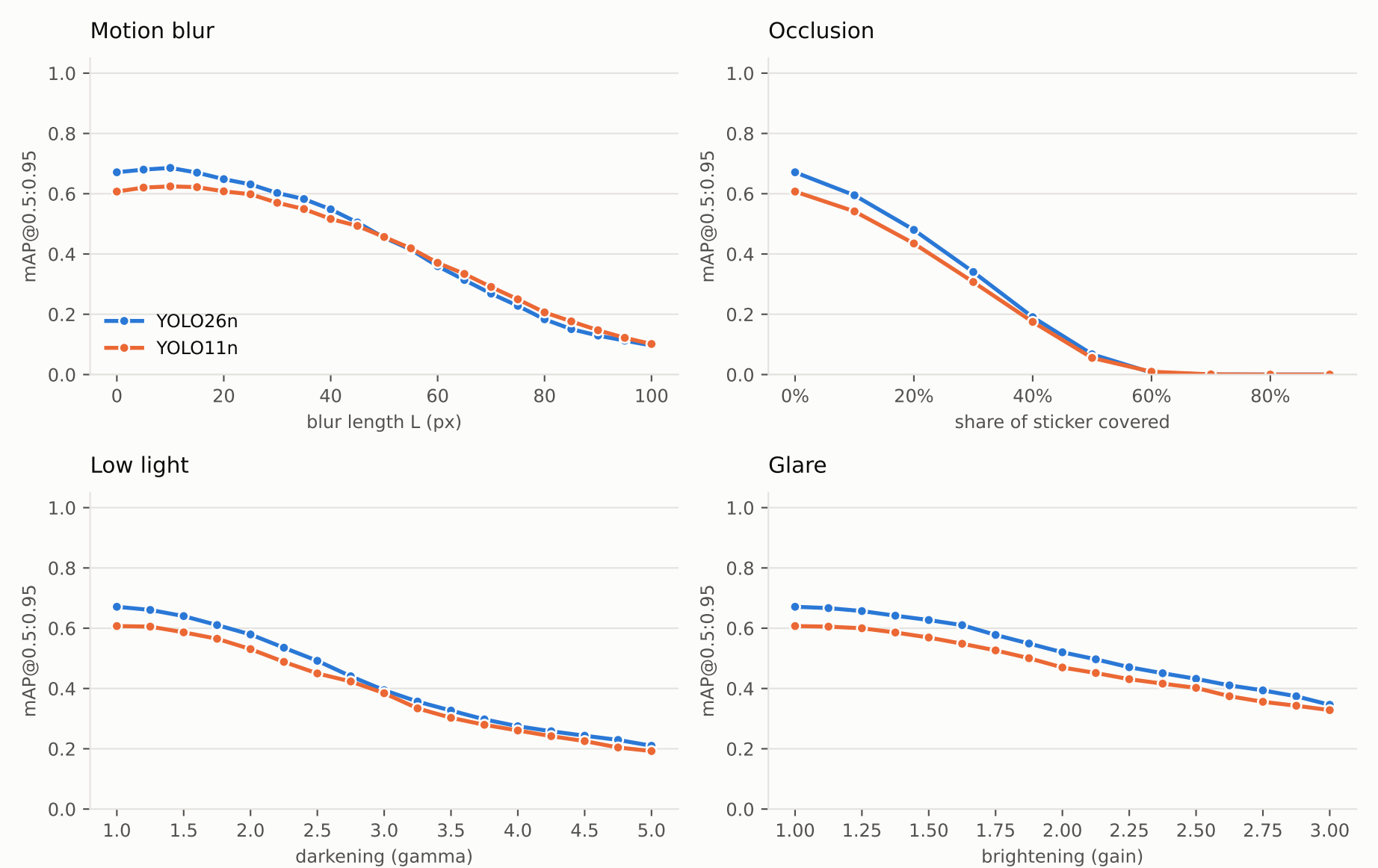


Bar shades, light to dark to medium: prepare image, model, cleanup (NMS). YOLO26n's cleanup is shorter because it is NMS-free.

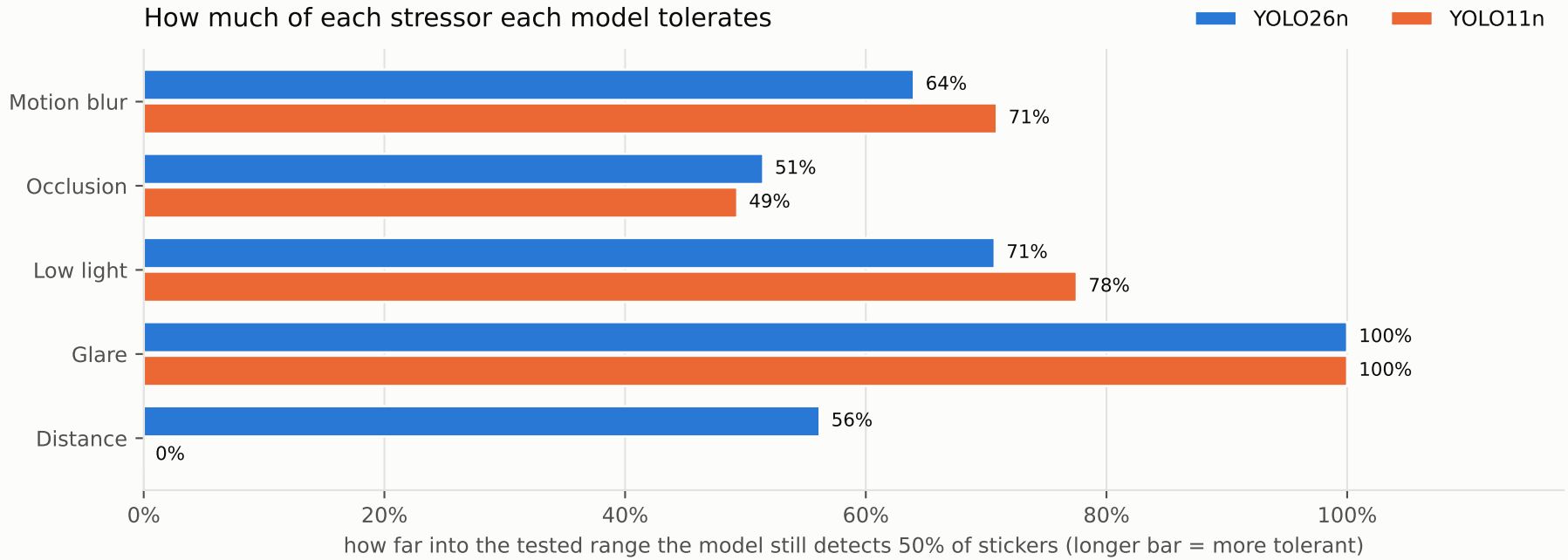

Detection rate by distance
share of stickers detected
1.0
0.8
0.6
0.4
0.2
0.0
reliable 90%
breaking point 50%
YOLO26n measured
YOLO26n fitted
YOLO11n measured
YOLO11n fitted
6.0
6.5
7.0
7.5
8.0
8.5
9.0
9.5
10.0
distance from camera (m)
Accuracy (mAP@0.5:0.95) by distance
mAP@0.5:0.95
YOLO26n
YOLO11n
distance from camera (m)



Dots = measured, lines = fitted curve. Dashed line = breaking point (50% detected); dotted line = reliable detection (90%). Further right = harder.



Further right = harder.


How much of each stressor each model tolerates
YOLO26n
YOLO11n
Motion blur
64%
71%
Occlusion
51%
49%
Low light
71%
78%
Glare
100%
100%
Distance
56%
0%
0%
20%
40%
60%
80%
100%
how far into the tested range the model still detects 50% of stickers (longer bar = more tolerant)